

Overview

Motivation

Archer fish can leap from the water to catch airborne prey (Figure 1). Notably, they produce sustained thrust throughout the duration of water exit, even when only partially submerged, and require little space for acceleration. This behavior serves as inspiration as a non-invasive alternative to existing water-exit strategies for aerial-aquatic vehicles [1].

Approach

We manipulated a mechanical fin model to reproduce the flapping behavior of archer fish locomotion. The mechanical model accommodates varied flapping kinematics such as stroke amplitude, frequency, and fin submergence depth. Force data collected during fin motion at fixed depths were analyzed to determine what effect flapping parameters, along with fin stiffness, have on thrust production. We used these results to design flapping trajectories for sustained thrust production during water exit. We also measured flow patterns and surface deformations through high-speed imaging to understand the loads produced.



Figure 1. An archer fish crossing the air-water interface (from [2]).

Experimental Methods

Mechanism

The experiment mechanism (Figure 2) actuates repeatable two-axis trajectories. Heave motion flaps the fin leading edge in the horizontal axis, while tow motion moves the fin in the vertical axis. The system allows the adjustment of five variables: heave amplitude, heave frequency, tow speed, fin submergence depth, and fin stiffness.

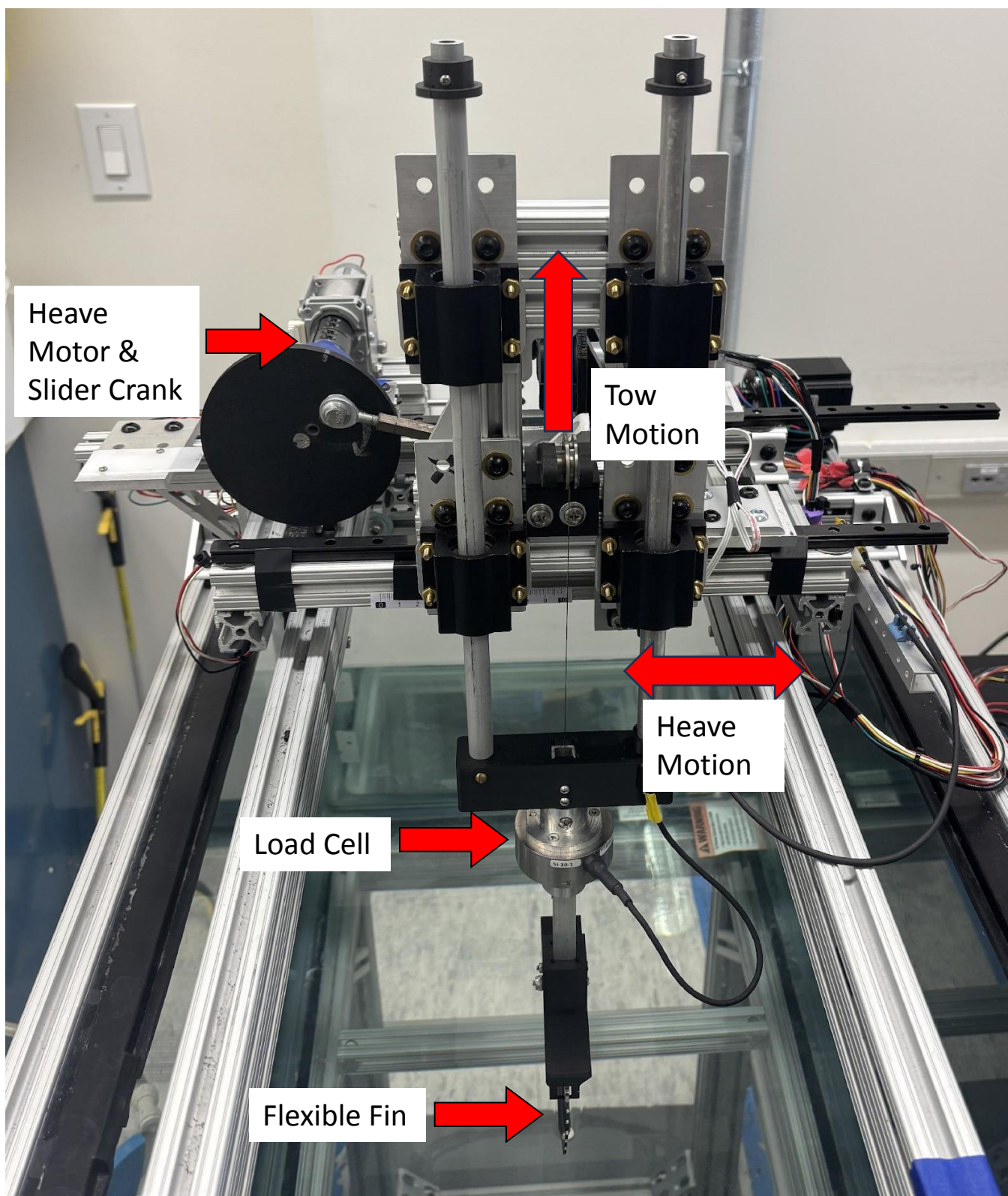


Figure 2: Tank and mechanism design

Force Measurements

A six-axis load cell measured forces and torques experienced by the fin. Each trajectory was also tested without water in the tank to capture inertial loads. The inertial loads were then subtracted from experimental measurements, isolating the forces solely from the fin's interaction with the fluid.

PIV Imaging

Particle Image Velocimetry (PIV) was used to image fin wake patterns (Figure 3). Suspended 55µm polyamide particles were illuminated by a 532 nm laser, and their movement was captured by a 500-fps camera. Points on the fin were illuminated with red light, allowing separate visual analysis of the red fin and green particles. PIV results allow us to calculate vorticity and other wake properties for comparison to the force measurements.

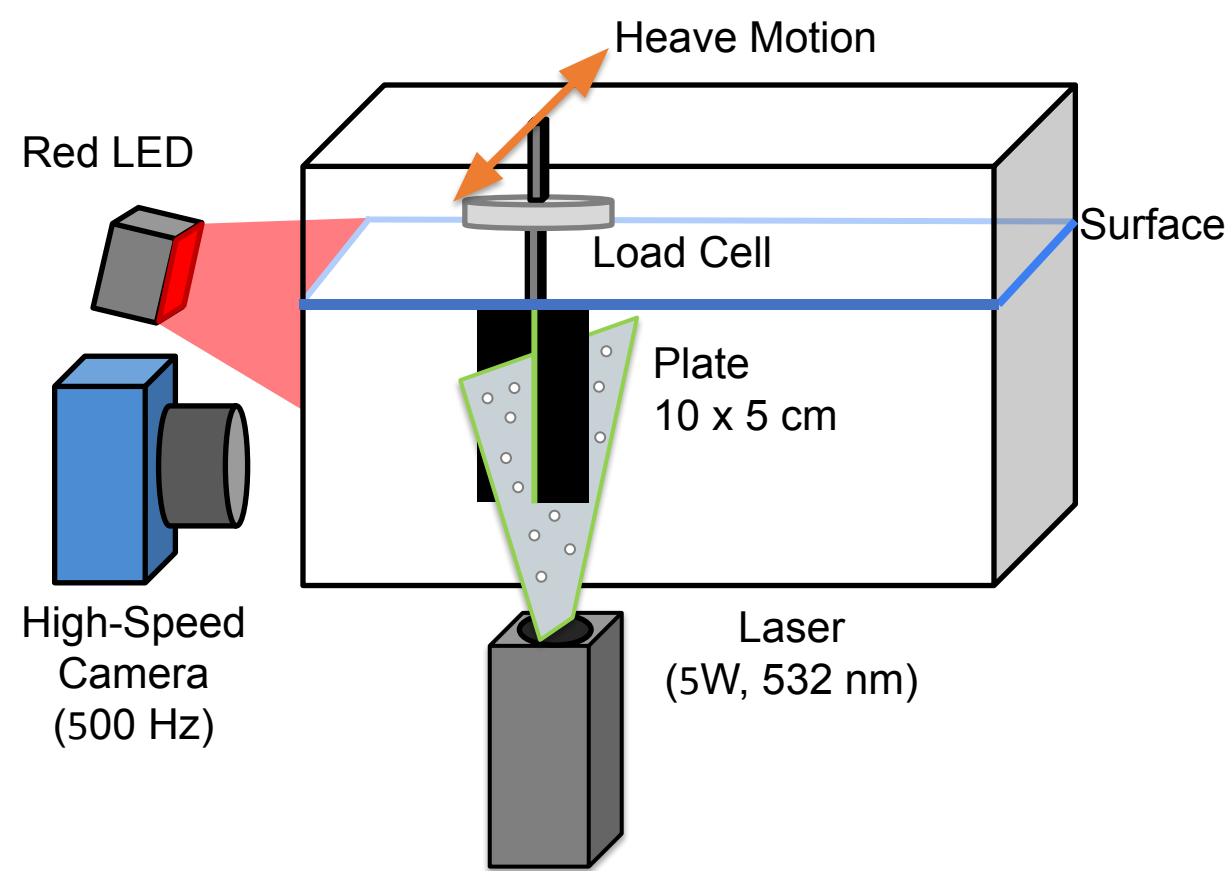


Figure 3: Experimental setup for PIV

Data Collection

The fixed depth parameter space for trajectory design spanned flapping amplitude A , flapping frequency f , plate stiffness (via thickness) and submergence, and vertical tow velocity U (Table 1). Five replicate trials were conducted for each set of conditions. Synchronous force measurements and PIV data were collected simultaneously. The measured fin thrust can be mapped to the vortex dynamics in its wake as measured by PIV.

	1, 2, 3 (cm)
	1, 2, 3 (Hz)
Fraction Submerged	25, 50, 75, 100 (%)
Plate Thickness	1/32", 1/16", 3/32", 1/8"

Table 1: Fixed depth data collection parameter space

Performance Assessment

To assess propulsive performance, the measured force for each fixed-depth trial was time-averaged over each peak-to-peak stroke to yield the mean thrust, $\bar{T}$, where

$$\bar{T} = \frac{1}{\tau} \int_0^{\tau} F_y(\tau) d\tau.$$

The mean thrust was nondimensionalized to calculate a thrust coefficient, C_T , used to assess the performance across different parameters. The characteristic velocity for the thrust coefficient was determined from the stroke peak-to-peak width $2A$ and the stroke frequency f . We also considered the thrust to power input ratio, ϵ , which compares the thrust output to the average power input, as a metric for efficiency. Kinematics were non-dimensionalized using the stroke-based Froude number (Fr) to account for the size and speed of the free surface disturbance. Figure 4 summarizes the collected fixed-depth data used to construct trajectories.

$$C_T = \frac{\bar{T}}{4\rho b L f^2 A^2} \quad \epsilon = \frac{\bar{T}}{\frac{1}{\tau} \int_0^{\tau} F_x(\tau) U_x(\tau) d\tau} \quad Fr = f \sqrt{\frac{2A}{g}}$$

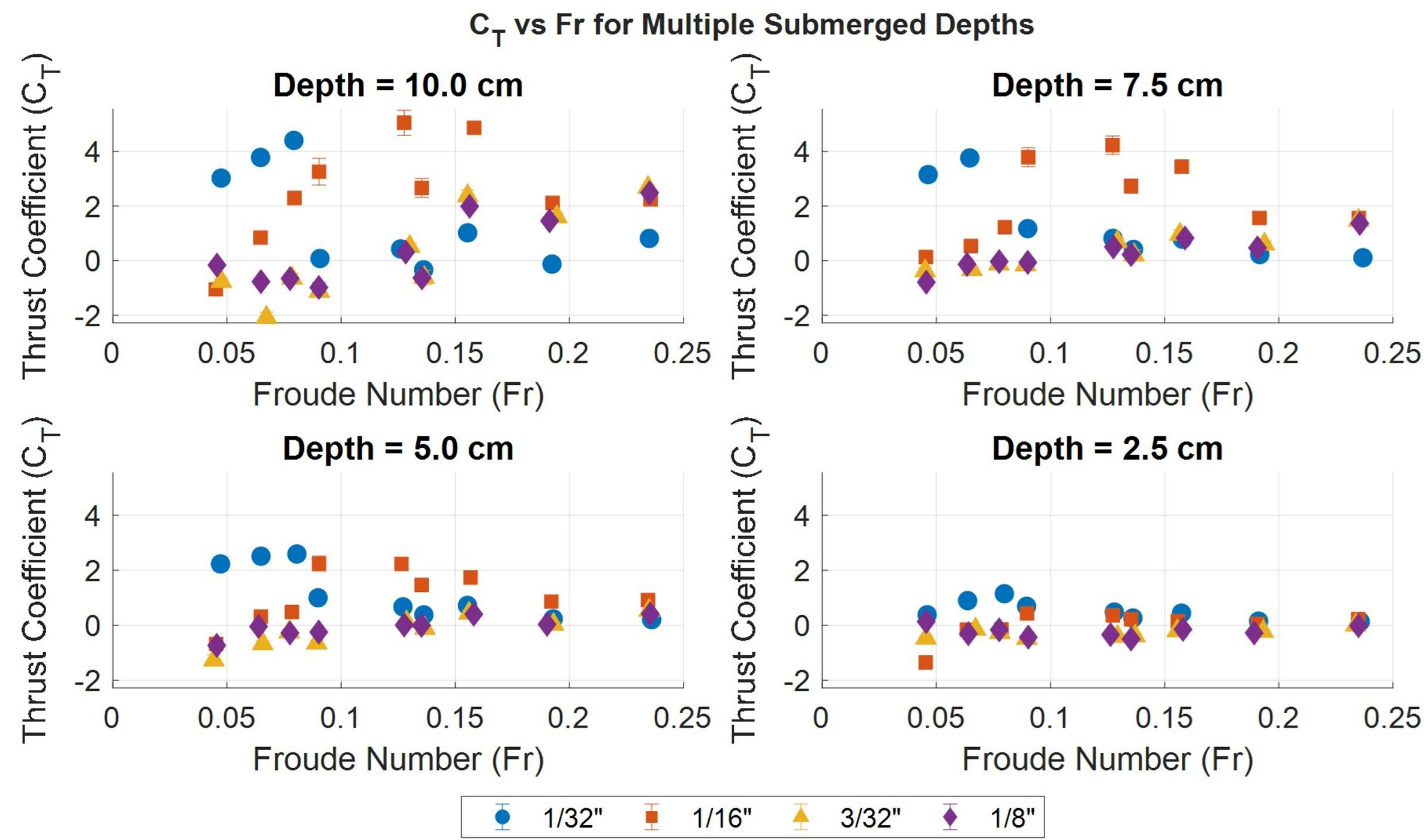


Figure 4: Fixed-Depth Thrust Results

Wake Results

Phase-averaged vorticity fields measured by PIV show the wake patterns produced in different parameter sets and allow comparison with force measurements. Figure 4 shows wake topology for three cases that all generate positive thrust at different depths, stiffnesses, and frequencies. The 25% submerged result (left), while barely thrust positive, also demonstrates that wake features can be resolved clearly even in the near-surface region.

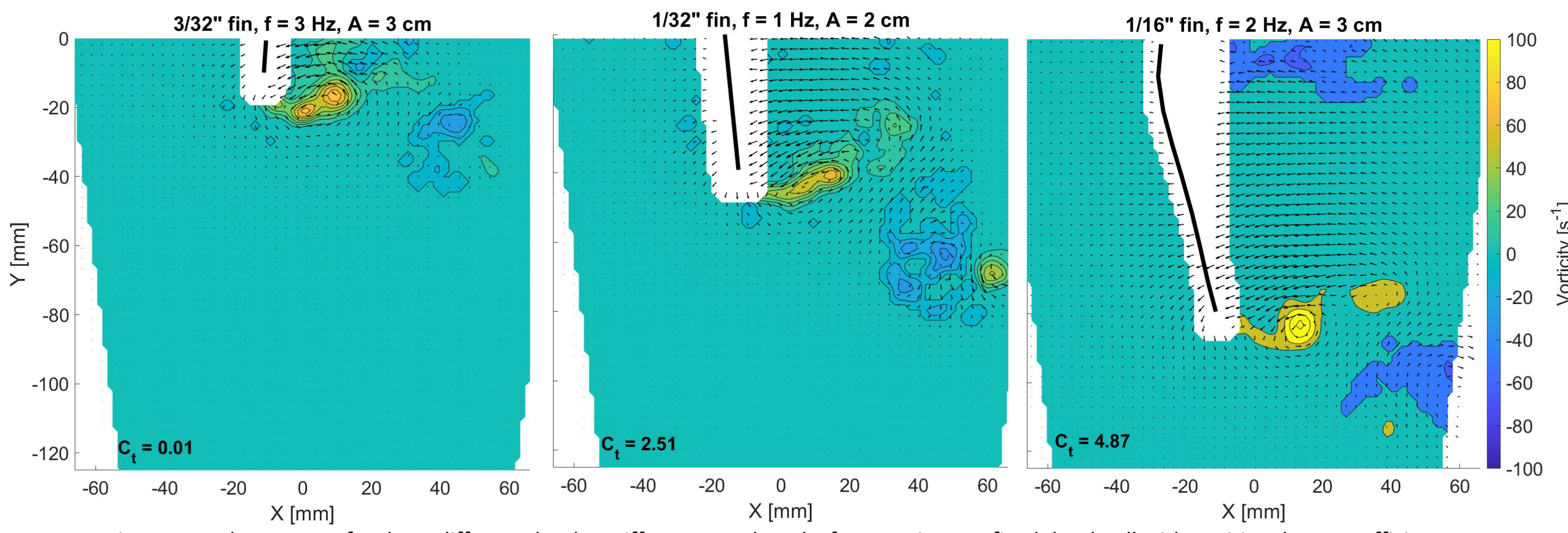


Figure 5: Wake patterns for three different depths, stiffnesses, and stroke frequencies at a fixed depth, all with positive thrust coefficients

Trajectory Planning and Testing

Trajectories were designed to maximize three parameters from flapping onset to water exit in towed cases: stroke-averaged thrust, thrust coefficient, and thrust to power input ratio. While plate stiffness and tow speed remained constant within a trajectory, new amplitude and frequency values could be chosen for each stroke to maximize one of the three metrics for the instantaneous depth.

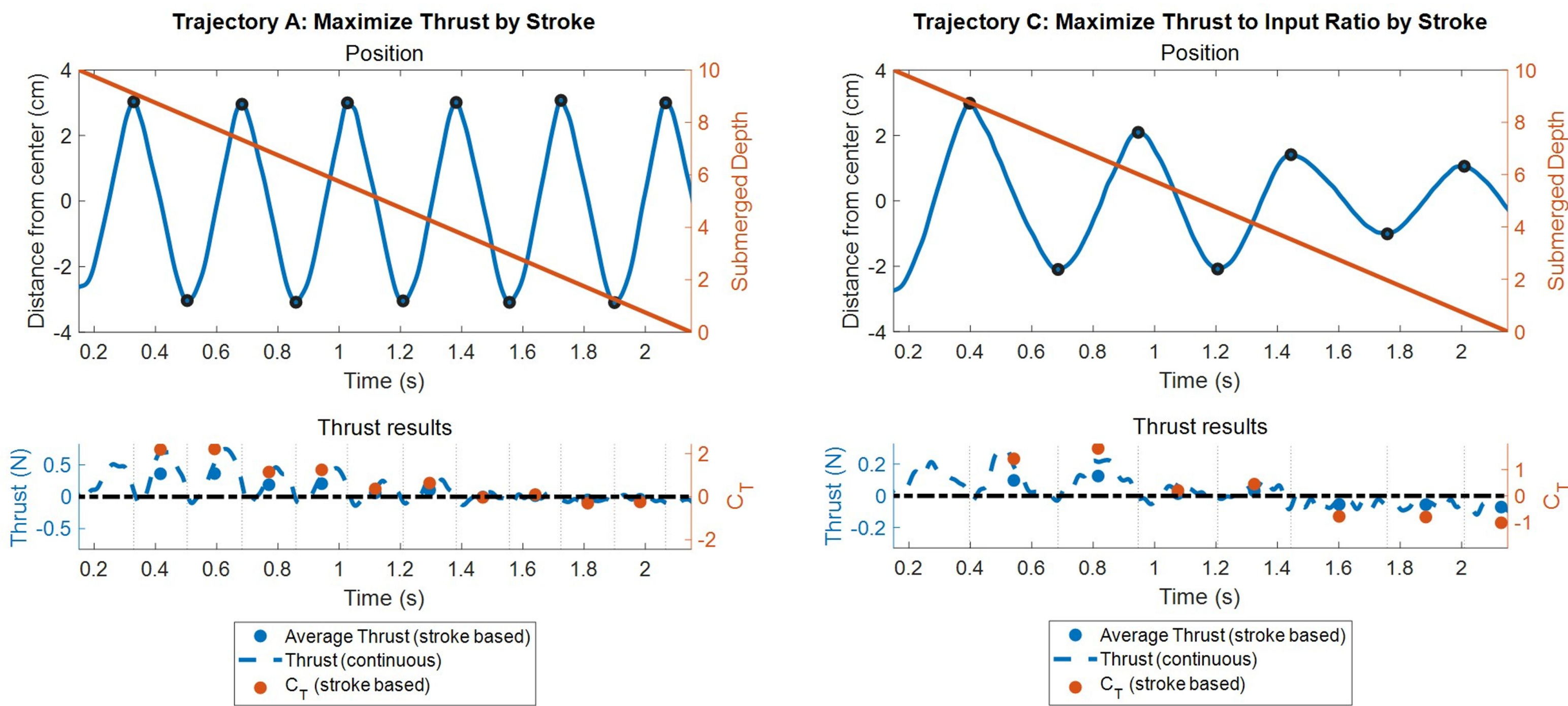


Figure 6: Sample 1/16" fin trajectory results

Figure 6 shows the position and thrust results for trajectories A and C, which were both designed for a 1/16" fin with a tow speed of 5 cm/s. Trajectory A optimized for absolute thrust with a faster and higher amplitude movement profile, while trajectory C optimized for thrust to power input ratio. While trajectory C produced less cumulative thrust, it did so more efficiently than trajectory A.

Conclusion and Future Work

Using fixed depth results, we were able to develop flapping trajectories capable of meaningful thrust production while exiting the water. Preliminary results show there is significant variety in the propulsion produced between parameter sets, but that these variable amplitude and frequency trajectories are largely more effective and efficient than fixed-parameter options. Additionally, these results show how fixed-depth results can be used to explain variable-depth thrust performance. Further exploration is necessary to deduce specific patterns and characteristics of propulsive performance.

Experimentation with trajectory design will be continued, with development of trajectories not simply optimizing thrust at each stroke, but the cumulative thrust across all strokes. Further investigation into surface deformations and air cavities, as well as using PIV to understand vorticity and wake patterns, will continue to refine propulsion strategies and techniques.

References & Acknowledgements

- [1] Sidall, Kovak (2017), *IEEE/ASME Transactions on Mechatronics*
 [2] Shih, Mendelson, Techet (2017) *J. Exp. Bio.*



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